Supply chain data cleaning

A Python pipeline that turns a messy shipments file into an analysis-ready table, with every cleaning step logged and checked.

The data in this project is simulated. I built a raw file with the kinds of problems real supply chain exports have, so the cleaning steps could be developed and tested on a known problem. The suppliers are made up and the emission factors are illustrative, not official figures.

The problem

The raw file (`raw_shipments.csv` , 570 rows) has these issues:

- 50 duplicate rows, some exact copies and some re-typed with different capitalisation and spacing
- Supplier names typed several ways (case, extra spaces, “Ltd” vs “Limited”, “Co” vs “Company”)
- Country names mixed with codes (“Nigeria”, “nigeria”, “NG”)
- Transport mode labels that mean the same thing (“Truck” and “Road”, “Ship” and “Sea”)
- Four different date formats, plus blanks and “TBC”
- Distances written as “1,240 km”, with some negative or impossible values
- Weights in tonnes or kilograms, with “N/A” and “unknown” for missing
- Costs with “$” signs and thousands separators
- Emissions missing for about 30 percent of shipments

What the pipeline does

1. Standardise shipment IDs and remove duplicate rows (570 rows down to 520)
2. Standardise supplier names so each supplier appears one way
3. Map country and transport mode labels to one standard form
4. Parse the four date formats into one (slashes are read as day/month/year)

5. Convert distance, weight and cost to numbers, and convert kilograms to tonnes
6. Set impossible values (negative or very large distances) to missing
7. Fill missing distances with the median for the same supplier and mode, and flag those rows
8. Estimate missing emissions as distance x weight x emission factor, and flag those rows
9. Run checks (unique IDs, no unmapped labels, exactly seven suppliers) and save the outputs

`cleaning_log.csv` records how many rows and cells each step changed.

Results

	Raw file	After cleaning
Rows	570	520
Missing distance	5.6%	0% (55 filled and flagged)
Missing emissions	30.7%	1.9% (146 estimated and flagged)
Supplier names	many spellings	7

Weight, cost and ship date still have gaps (about 3 to 6 percent each). I left them missing on purpose, because guessing a weight or a date would add error without good evidence.

A quick analysis on the clean data shows air freight carries 15% of shipments but about 77% of the emissions, while road carries half the shipments and about 14%. In a real project this is where the cleaned data starts to answer a question.

Decisions and limits

- **Imputed values are flagged.** The columns `distance_imputed` and `emissions_estimated` mark every filled value, so anyone using the data can exclude them.
- **Emissions to review.** 25 rows where the reported emissions differ from the calculated value by more than 30 percent are marked `review`. Many of these are rows where the distance had to be filled in.
- **Date format assumption.** Dates with slashes are treated as day/month/year. With real

data I would confirm this with whoever supplied the file.

- **Simulated data.** The pipeline is real, but the numbers say nothing about real suppliers.

Files

File	What it is
`simulate_raw_data.py`	Builds the messy raw file
`clean_supply_chain_data.py`	The cleaning pipeline
`raw_shipments.csv`	Raw simulated input
`clean_shipments.csv`	Cleaned output
`cleaning_log.csv`	What each step changed
`data_quality.png`	Before and after chart, and emissions by mode

Run it

```
pip install numpy pandas matplotlib
python simulate_raw_data.py
python clean_supply_chain_data.py
```

The random seed is fixed, so you get the same numbers every time.